

SUMMARY

I am an applied economist with a Ph.D. in energy and environmental economics and 9+ years of experience delivering economic analysis for governments, development finance institutions and private-sector clients. My work focuses on the economics of energy markets, regulation and infrastructure investment, with a particular emphasis on translating complex technical and policy questions into clear economic evidence for decision-makers. I combine bottom-up financial and market modelling with econometric and macroeconomic analysis to answer questions about fuel costs and dispatch, tariff structures, financing and cost of capital, and the wider economic value of reliable energy supply. Across the World Bank Group, McKinsey & Company and independent advisory work, I have delivered rigorous economic analysis for senior government, financial-sector and institutional audiences across Europe, Central Asia, Latin America and the Middle East.

EXPERIENCE

Regovia, London, UK

July 2026 – Present

Founder and Principal

Independent economics advisory practice focused on energy and power-sector economics, climate and environmental regulation, and economic policy appraisal.

- Developed and presented analysis for Malaysia's Ministry of Energy Transition and Water Transformation (PETRA) on international approaches to electricity-grid financing and public-private partnerships.
- For Brazil's Ministry of Finance, analysed the cost pass-through of the proposed emissions trading system in electricity tariffs, and the placement of compliance obligations in the electricity supply chain

The World Bank Group, London, UK

October 2024 – Present

Consultant, Climate Change Group - Finance and Economics Team

- Originated the design of a World Bank-supported blended-finance programme of up to USD 450 million with TSKB, Türkiye's industrial development bank, including the analysis of its renewable-energy portfolio, investment pipeline and financing structure.
- Designed and delivered macroeconomic impact analysis of climate policies using WBG-IMF CGE modelling tools for government clients in the ECA region, quantifying effects on fiscal impact and industrial competitiveness.
- Developed a bottom-up energy investment model for a combined grant in Türkiye covering more than 61,000 coal-heated homes, estimating heating demand, technology sizing, grid requirements, capital and operating costs, household payback and the fiscal cost of alternative financing structures.
- Co-developed the World Bank's methodology for global sectoral analysis of environmental and energy taxation with the lead economist in the World Bank's Planet Research Team.

McKinsey & Company, London, UK

June 2022 – October 2024

Senior economist (Associate consultant – junior engagement manager equivalent)

- Built NPV, fuel-demand and greenhouse-gas abatement models for zero-emission maritime fuels and supported the commercial and tender design of a zero-emission shipping procurement programme, including analysis of alternative fuel costs and regulatory scenarios.
- Led quantitative impact analysis of environmental regulation on residential and manufacturing sector energy demand for major US states to inform state-level climate policy design.
- Served as methodology lead for climate risk assessment projects with a major European IFI, designing quantitative modelling frameworks combining physical and transition risk analysis for infrastructure investment decisions.
- Led data analytics for portfolio decarbonisation initiatives of leading European and UK financial institutions including coal-fired power generation and renewable energy.
- Advised government agencies in Gulf states on government-led private sector investment programmes for low-carbon economy transition in renewable energy, sustainable fuels and green manufacturing.
- Facilitated C-suite executive workshops on identifying green business opportunities for corporate and investment banks.

The World Bank Group, Washington D.C., USA
Consultant, Carbon Pricing and Innovation Practice

July 2019 – May 2022

- Lead analytical contributor to the delivery of State and Trends of Carbon Pricing reports (2021, 2022) and manager of the Carbon Pricing Dashboard, facilitating quantitative analysis and evidence-based policy research supporting governments and international organizations.

Tufts University, Massachusetts, USA
Research associate

July 2017 – May 2022

- Estimated macroeconomic impacts of environmental regulations in North America and Europe with Gilbert Metcalf and James Stock (Harvard University).
- Conducted agent-based modelling in MATLAB to study general equilibrium effects of microcredit initiatives and performed non-parametric econometric analysis on informal insurance in developing countries for 2019 Nobel Prize winners in economics Esther Duflo (MIT), and Abhijit Banerjee (MIT).
- Analysed agricultural productivity dispersion using microlevel household data with R and Python in an NSF-funded project for Kyle Emerick and Marshall Burke (Stanford University).

EDUCATION

Tufts University, Massachusetts, USA
Ph.D. in Economics, concentration on environmental and energy economics

September 2017 – May 2022

New York University, New York, USA
M.A. in Political Economy

September 2013 – June 2015

Bogazici University, Turkey
B.A. in Political Science

December 2010

ACADEMIC ECONOMICS RESEARCH

- “How does the increased share of renewable energy impact electricity markets and climate?: A natural experiment using the COVID-19 negative demand shock.” ([Link](#))
 - Job market paper using approximately 39,000 hourly observations from Türkiye's electricity market to estimate the effects of renewable generation on wholesale electricity prices, thermal dispatch, fuel displacement and emissions.
- “Asymmetric price elasticity of supply and drilling rates in shale oil.” (with Michi Toya) ([Link](#))
 - Econometric research on the price elasticity of US shale oil supply and drilling activity, analysing how production responds to energy-price changes under geological, technological and regulatory constraints.
- “Green Taxes and Yellow Vests: The effect of the French carbon tax on capital market responses and firm performance.” ([Link](#))
- “Price shocks, liquidity constraints and agricultural outcomes: The case of coffee production in Ethiopia.” ([Link](#))

SKILLS

Programming languages and software: Python, R, Stata, MATLAB, SQL, ArcGIS, Google Earth Engine, Qualtrics.

Economic and climate modelling: Econometric methods (causal inference, quasi-experimental designs), climate risk assessment, CGE modelling (WBG-IMF CPAT), energy system modelling (LEAP, NEMO, OSeMOSYS).

Languages: Turkish (native), English (native equivalent), French (B1)